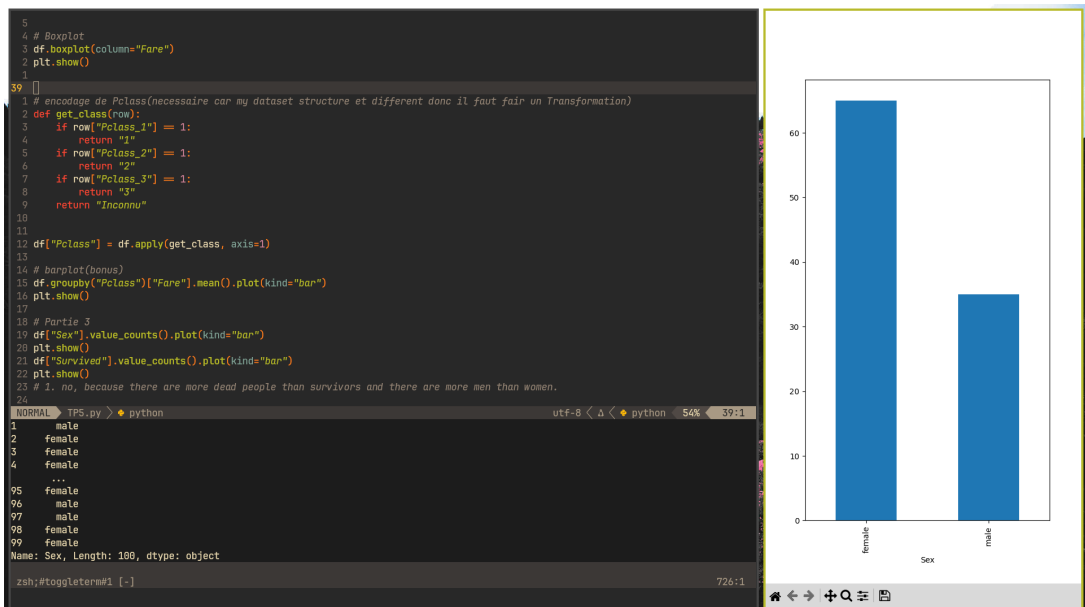
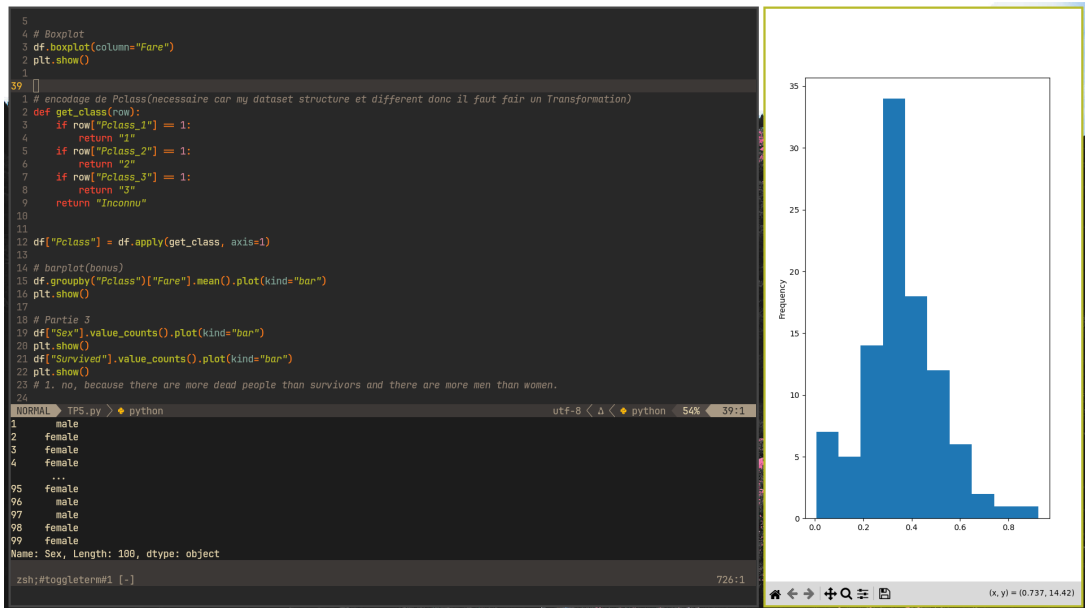


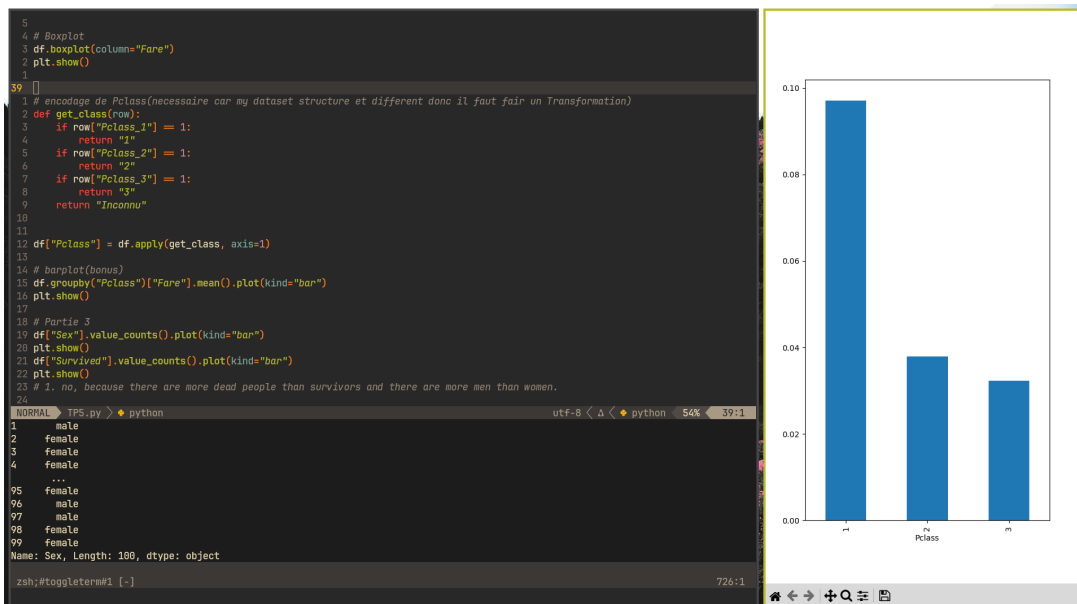
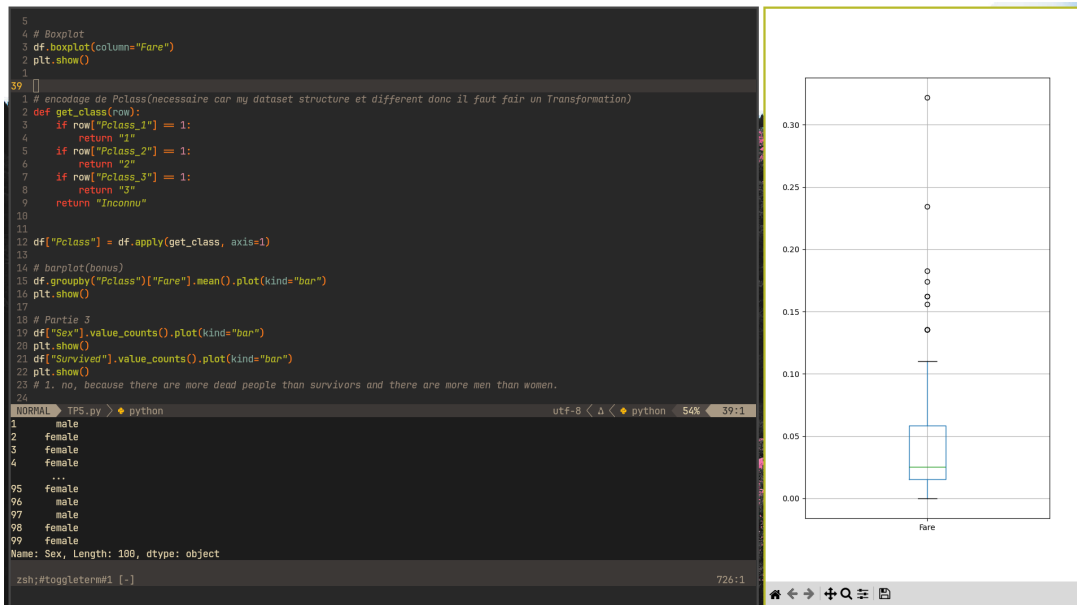
TP5

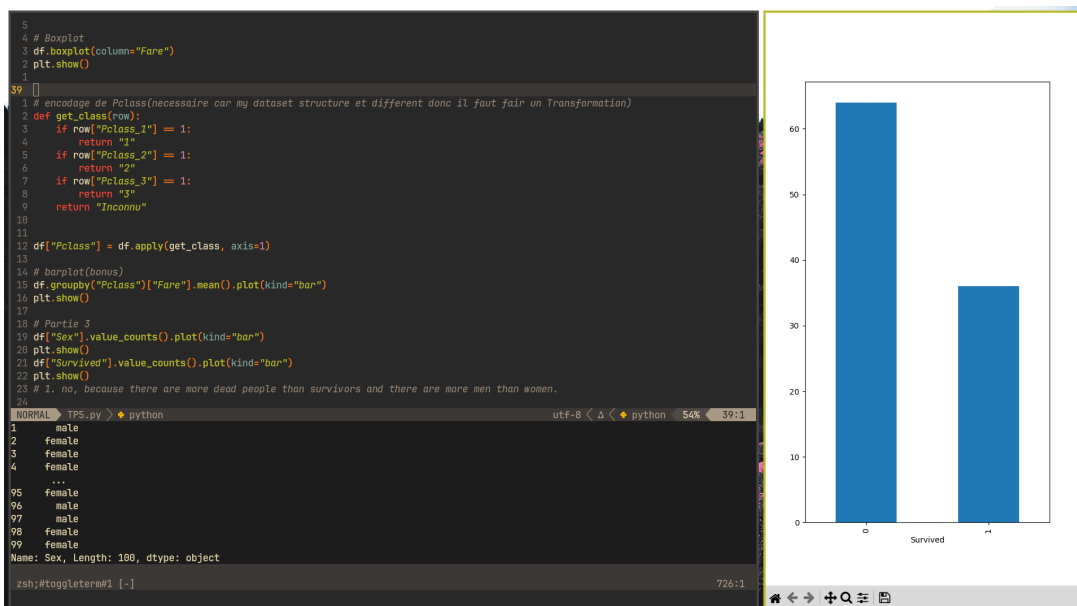
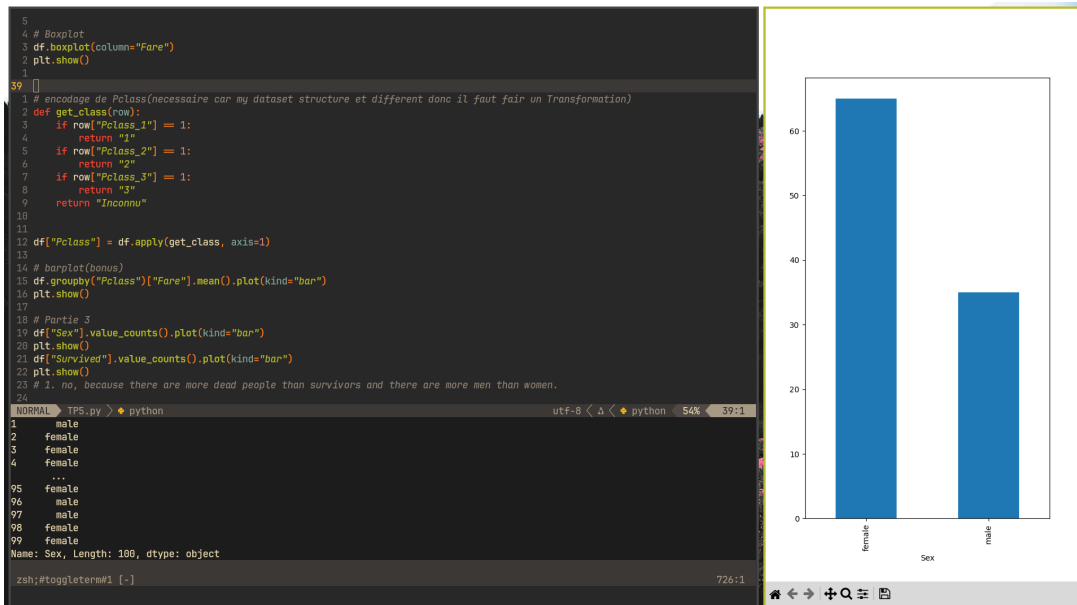
Mohamed Ali Knis - TD3/TP6

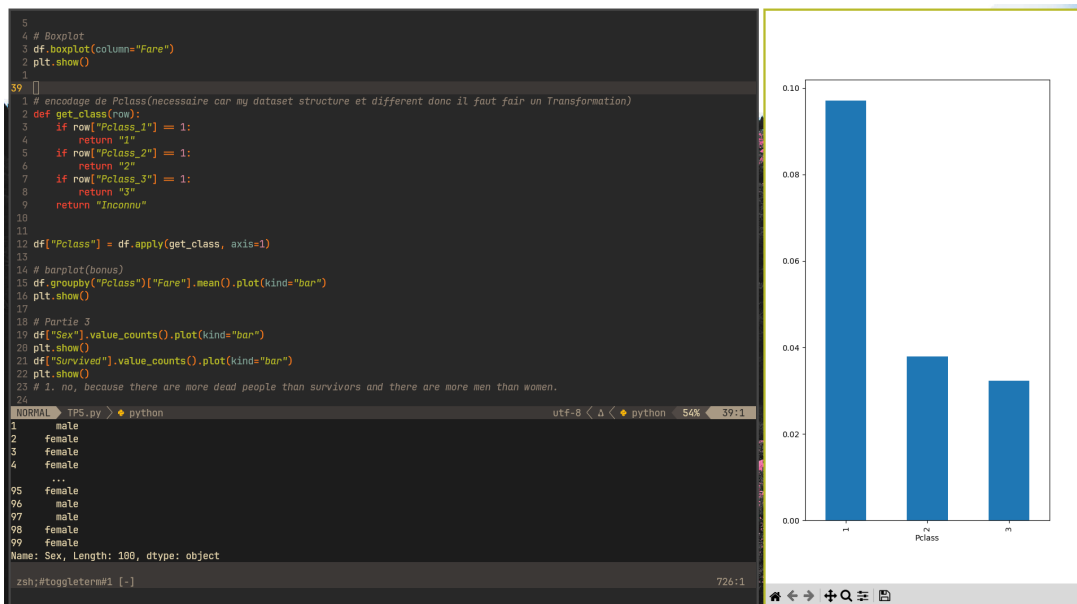
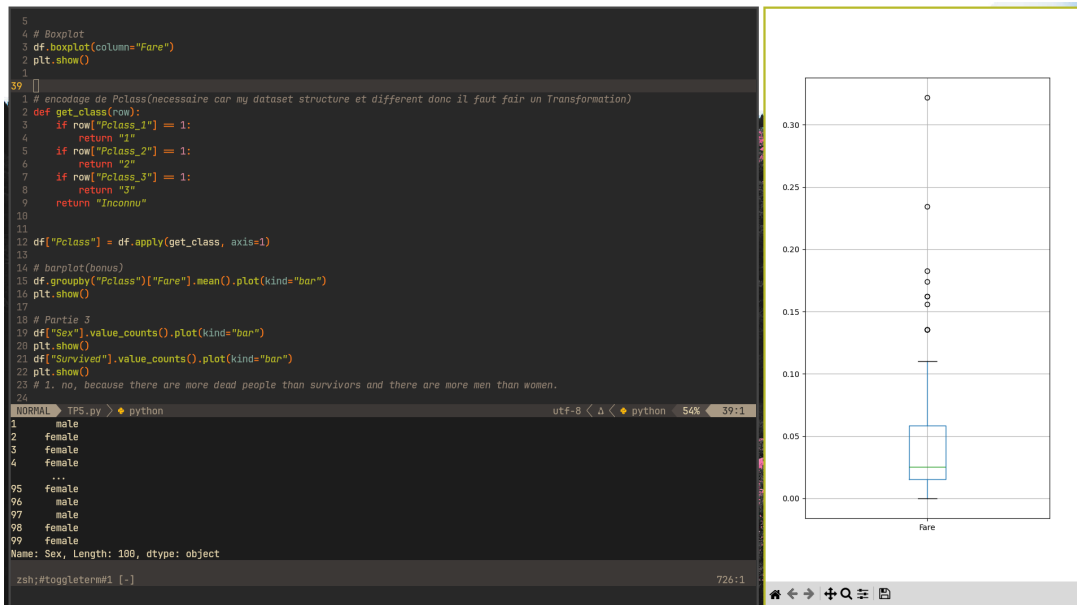
```
37 import kagglehub
38 from kagglehub import KaggleDatasetAdapter
39 import matplotlib.pyplot as plt
40
41 df = Kagglehub.dataset_load(
42     KaggleDatasetAdapter.PANDAS,
43     "azeembootwala/titanic",
44     "test_data.csv",
45 )
46
47 # Partie 1
48 print(df.head())
49 print(df.shape)
50 print(df.isnull().sum())
51
52 print(df["Sex"])
53 df["Age"] = df["Age"].fillna(df["Age"].mean())
54 print(df["Sex"])
55
56 print(df["Sex"])
57 df["Sex"] = df["Sex"].map({0: "male", 1: "female"})
58 print(df["Sex"])
59
60 # Partie 2
61
62 # histogramme
63 df["Age"].plot(kind="hist")
64 plt.show()
65
66 # barplot
67 df["Sex"].value_counts().plot(kind="bar")
68 plt.show()
69
70 # Boxplot
71 df.boxplot(column="Fare")
72 plt.show()
73
74 # encodage de Pclass (nécessaire car my dataset structure est différent donc il faut faire une transformation)
75 def get_class(row):
76     if row["Pclass_1"] == 1:
77         return "1"
78     if row["Pclass_2"] == 1:
79         return "2"
80     if row["Pclass_3"] == 1:
81         return "3"
82     return "Inconnu"
83
84 df["Pclass"] = df.apply(get_class, axis=1)
85
86 # barplot (bonus)
87 df.groupby("Pclass")["Fare"].mean().plot(kind="bar")
88 plt.show()
89
90 # Partie 3
91 df["Sex"].value_counts().plot(kind="bar")
92 plt.show()
93 df["Survived"].value_counts().plot(kind="bar")
94 plt.show()
95
96 # 1. non, car il y a plus de morts que de survivants et il y a plus d'hommes que de femmes.
97
98 df.boxplot(column="Fare")
99 plt.show()
100
101 # 2. oui, car le boxplot montre les valeurs extrêmes
102
103 df.groupby("Pclass")["Fare"].mean().plot(kind="bar")
104 plt.show()
105 print(df.groupby("Pclass")["Fare"].mean())
106
107 # 3. la classe 1 paie le plus avec un coût moyen plus que la somme des deux autres classes combinées.
```

```
5
6 # Boxplot
7 df.boxplot(column="Fare")
8 plt.show()
9
10 # encodage de Pclass (nécessaire car my dataset structure est différent donc il faut faire une transformation)
11 def get_class(row):
12     if row["Pclass_1"] == 1:
13         return "1"
14     if row["Pclass_2"] == 1:
15         return "2"
16     if row["Pclass_3"] == 1:
17         return "3"
18     return "Inconnu"
19
20 df["Pclass"] = df.apply(get_class, axis=1)
21
22 # barplot (bonus)
23 df.groupby("Pclass")["Fare"].mean().plot(kind="bar")
24 plt.show()
25
26 # Partie 3
27 df["Sex"].value_counts().plot(kind="bar")
28 plt.show()
29 df["Survived"].value_counts().plot(kind="bar")
30 plt.show()
31
32 # 1. non, car il y a plus de morts que de survivants et il y a plus d'hommes que de femmes.
33
34 df.boxplot(column="Fare")
35 plt.show()
36
37 # 2. oui, car le boxplot montre les valeurs extrêmes
38
39 df.groupby("Pclass")["Fare"].mean().plot(kind="bar")
40 plt.show()
41 print(df.groupby("Pclass")["Fare"].mean())
42
43 # 3. la classe 1 paie le plus avec un coût moyen plus que la somme des deux autres classes combinées.
```









```
3
4 # Boxplot
5 df.boxplot(column="Fare")
6 plt.show()
7
39
1 # encodage de Pclass (nécessaire car my dataset structure est différent donc il faut faire une transformation)
2 def get_class(row):
3     if row["Pclass_1"] == 1:
4         return "1"
5     if row["Pclass_2"] == 1:
6         return "2"
7     if row["Pclass_3"] == 1:
8         return "3"
9     return "Inconnu"
10
11
12 df["Pclass"] = df.apply(get_class, axis=1)
13
14 # barplot (bonus)
15 df.groupby("Pclass")["Fare"].mean().plot(kind="bar")
16 plt.show()
17
18 # Partie 3
19 df["Sex"].value_counts().plot(kind="bar")
20 plt.show()
21 df["Survived"].value_counts().plot(kind="bar")
22 plt.show()
23 # 1. no, because there are more dead people than survivors and there are more men than women.
24
NORMAL TPS.py > python utf-8 < python 54% 39:1
97 male
98 female
99 female
Name: Sex, Length: 100, dtype: object
Pclass
1 0.697096
2 0.837951
3 0.832388
Name: Fare, dtype: float64
Dev/BI/TPS via v3.13.12 via impure (nix-shell-env) took 16s
zsh:toggleterm#1 [-] 733:1
```